

Minimization of Deterministic Finite Automata

Definitions, Algorithm, and Proofs

Tarun Hariharan

August 6, 2026

1 Problem Context

Let $M = (Q, \Sigma, \delta, q_0, F)$ be a Deterministic Finite Automaton (DFA) recognizing the language $L(M)$.

- Q is the finite set of states.
- Σ is the input alphabet.
- $\delta : Q \times \Sigma \rightarrow Q$ is the transition function.
- $q_0 \in Q$ is the start state.
- $F \subseteq Q$ is the set of accepting states.

We assume without loss of generality that all states in Q are reachable from q_0 . Let $\hat{\delta} : Q \times \Sigma^* \rightarrow Q$ denote the extended transition function.

2 Core Definitions

Definition 1 (Indistinguishability $\equiv$). Two states $p, q \in Q$ are indistinguishable, denoted as $p \equiv q$, if and only if for all strings $w \in \Sigma^*$:

$$\hat{\delta}(p, w) \in F \iff \hat{\delta}(q, w) \in F$$

If such a string w exists where one transitions to an accept state and the other does not, w is said to distinguish p and q ($p \not\equiv q$). The relation $\equiv$ is an equivalence relation on Q (can be shown easily and is a good exercise to show).

Definition 2 (k -Indistinguishability $\equiv_k$). Two states $p, q \in Q$ are k -indistinguishable, denoted as $p \equiv_k q$, if and only if for all strings $w \in \Sigma^*$ where $|w| \leq k$:

$$\hat{\delta}(p, w) \in F \iff \hat{\delta}(q, w) \in F$$

Two states are indistinguishable ($p \equiv q$) if and only if they are k -indistinguishable for all $k \geq 0$.

3 The Minimization Algorithm (Equivalence Class Method)

The algorithm systematically computes the equivalence relation $\equiv$ by iteratively computing $\equiv_k$ until the partitions stabilize.

1. **Initialization** ($k = 0$): Construct an initial partition P_0 of Q based on 0-indistinguishability.

$$P_0 = \{F, Q \setminus F\}$$

2. **Refinement** ($k \rightarrow k+1$): Given partition P_k , construct P_{k+1} . Two states p, q belong to the same block in P_{k+1} if and only if:
 - (a) $p \equiv_k q$ (they are in the same block in P_k)
 - (b) $\forall a \in \Sigma, \delta(p, a) \equiv_k \delta(q, a)$ (their transitions on all symbols go to the same block in P_k)
3. **Termination**: Stop when $P_{k+1} = P_k$. The blocks of this final partition are the equivalence classes under $\equiv$. Let $[q]$ denote the equivalence class containing state q .

We construct the minimized DFA $M_{\min} = (Q_{\min}, \Sigma, \delta_{\min}, q_{0,\min}, F_{\min})$ as follows:

- $Q_{\min} = \{[q] \mid q \in Q\}$
- $q_{0,\min} = [q_0]$
- $F_{\min} = \{[q] \mid q \in F\}$
- $\delta_{\min}([q], a) = [\delta(q, a)]$ for all $a \in \Sigma$

4 Proof of Correctness (Language Equivalence)

To prove $L(M_{\min}) = L(M)$, we must first establish that the constructed states and transitions are well-defined (independent of the chosen representative $q \in [q]$).

Lemma 1 (Well-definedness). *The definitions of $F_{\min}$ and $\delta_{\min}$ are well-defined.*

Proof. For $F_{\min}$: Suppose $p \equiv q$. Evaluating indistinguishability with $w = \epsilon$, we have $\hat{\delta}(p, \epsilon) \in F \iff \hat{\delta}(q, \epsilon) \in F$, which means $p \in F \iff q \in F$. Thus, an equivalence class contains either only accepting states or only rejecting states. $F_{\min}$ is well-defined.

For $\delta_{\min}$: We must show that if $p \equiv q$, then $\delta(p, a) \equiv \delta(q, a)$. Let $w \in \Sigma^*$ be arbitrary. Because $p \equiv q$, we evaluate the distinguishing string aw :

$$\hat{\delta}(p, aw) \in F \iff \hat{\delta}(q, aw) \in F$$

$$\hat{\delta}(\delta(p, a), w) \in F \iff \hat{\delta}(\delta(q, a), w) \in F$$

Since this holds for all w , $\delta(p, a) \equiv \delta(q, a)$. Thus, $\delta_{\min}$ is well-defined. □

Theorem 1. $M_{\min}$ recognizes the language $L(M)$.

Proof. We proceed by induction on the length of the string w to show that $\hat{\delta}_{\min}([q_0], w) = [\hat{\delta}(q_0, w)]$.

Base Case ($|w| = 0$): $w = \epsilon$.

$$\hat{\delta}_{\min}([q_0], \epsilon) = [q_0] = [\hat{\delta}(q_0, \epsilon)]$$

Inductive Step: Assume $\hat{\delta}_{\min}([q_0], w) = [\hat{\delta}(q_0, w)]$ holds for a string w . We prove it for wa , where $a \in \Sigma$.

$$\begin{aligned} \hat{\delta}_{\min}([q_0], wa) &= \delta_{\min}(\hat{\delta}_{\min}([q_0], w), a) \\ &= \delta_{\min}([\hat{\delta}(q_0, w)], a) \quad (\text{by inductive hypothesis}) \\ &= [\delta(\hat{\delta}(q_0, w), a)] \quad (\text{by definition of } \delta_{\min}) \\ &= [\hat{\delta}(q_0, wa)] \end{aligned}$$

Finally, we evaluate language acceptance for any $w \in \Sigma^*$:

$$\begin{aligned}
w \in L(M_{\min}) &\iff \hat{\delta}_{\min}([q_0], w) \in F_{\min} \\
&\iff [\hat{\delta}(q_0, w)] \in F_{\min} \\
&\iff \hat{\delta}(q_0, w) \in F \quad (\text{by definition of } F_{\min}) \\
&\iff w \in L(M)
\end{aligned}$$

Therefore, $L(M_{\min}) = L(M)$. □

5 Proof of Minimality

Theorem 2. *No DFA with fewer states than $M_{\min}$ can recognize $L(M)$.*

Proof. Assume, for the sake of contradiction, that there exists a DFA $M' = (Q', \Sigma, \delta', q'_0, F')$ such that $L(M') = L(M)$ and $|Q'| < |Q_{\min}|$. Assume all states in M' are reachable.

Because every equivalence class $[q] \in Q_{\min}$ contains states reachable from q_0 in M , for every class $[q]$ there exists a string $x_{[q]} \in \Sigma^*$ such that $\hat{\delta}(q_0, x_{[q]}) \in [q]$.

When we process these strings on the hypothetical machine M' , the number of strings equals $|Q_{\min}|$. Since $|Q'| < |Q_{\min}|$, by the **Pigeonhole Principle**, there must exist two distinct equivalence classes $[p] \neq [q]$ in $M_{\min}$ whose corresponding strings lead to the exact same state $r \in Q'$ in M' :

$$\hat{\delta}'(q'_0, x_{[p]}) = r = \hat{\delta}'(q'_0, x_{[q]})$$

Since $[p] \neq [q]$ in $M_{\min}$, the states p and q are distinguishable. Thus, there exists a string $w \in \Sigma^*$ that distinguishes them. Without loss of generality, assume:

$$x_{[p]}w \in L(M) \quad \text{and} \quad x_{[q]}w \notin L(M)$$

Now, observe how M' processes the strings $x_{[p]}w$ and $x_{[q]}w$:

$$\begin{aligned}
\hat{\delta}'(q'_0, x_{[p]}w) &= \hat{\delta}'(\hat{\delta}'(q'_0, x_{[p]}), w) = \hat{\delta}'(r, w) \\
\hat{\delta}'(q'_0, x_{[q]}w) &= \hat{\delta}'(\hat{\delta}'(q'_0, x_{[q]}), w) = \hat{\delta}'(r, w)
\end{aligned}$$

Because M' arrives at the identical state $\hat{\delta}'(r, w)$ for both strings, it must either accept both or reject both. However, $L(M)$ requires accepting exactly one. Thus, M' fails to recognize $L(M)$, which contradicts our initial assumption.

Therefore, no such smaller DFA M' exists, and $M_{\min}$ is minimal. □